FUN-4.B.1 The derivative can be used to solve optimization problems; that is, finding a minimum or maximum value of a function on a given interval.
FUN-4.C.1 Minimum and maximum values of a function take on specific meanings in applied contexts.

5-10 & 5-11 Optimization Problems

One of the most common applications of calculus involves the determination of minimum or maximum values. Consider how frequently you hear or read terms such as greatest profit, least cost, least time, greatest voltage, optimum size, least size, greatest strength and greatest distance.

Guidelines for Solving Applied Optimization Problems
(Maximum and Minimum problems)

1.) Identify all **given** quantities and quantities **to be determined.** If helpful, make a sketch
2.) Write a **primary equation** for the quantity to be maximized (or minimized).
3.) Reduce the primary equation to one having a single independent variable. This may require the use of a **secondary equation**.
4.) Determine any restrictions on all variables in the problem.
5.) Determine the desired maximum or (minimum) value by using the techniques discussed in Topics 5.4 & 5.5. This includes taking a derivative, finding critical numbers and setting up a chart.
6.) Be sure to state what was asked for in the problem!

Example 1: Find two positive numbers whose product is 100 and whose sum is a minimum.

Restrictions:

$x>0 \quad y>0$

Primary EQ: $S = x + y$

Secondary EQ: $x \cdot y = 100$

$x = \frac{100}{y}$

$S = x + y$

$S = \frac{100}{y} + y$

$S' = \frac{-100}{y^2} + 1 = \frac{-100 + y^2}{y^2}$

S': $-$ $+$

$0 \searrow 10 \nearrow$ min

$S' = 0$ when $-100 + y^2 = 0$

$y^2 = 100$

$y = \pm 10$

S' undef $\rightarrow y^2 = 0$

$y = 0$

The two numbers are 10 and 10

$y = 10$

$x = \frac{100}{10} = 10$

Example 2: A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Primary EQ: $A = L \cdot W$

Secondary EQ: $2400 = L + 2W$

$L = 2400 - 2W$

Restrictions:

$0 < L < 2400$

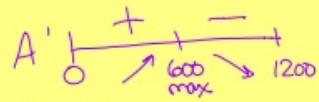
$0 < W < 1200$

$A = L \cdot W$

$A = (2400 - 2W)W$

$A = 2400W - 2W^2$

$A' = 2400 - 4W$



$2400 - 4W = 0$

$2400 = 4W$

$600 = W$

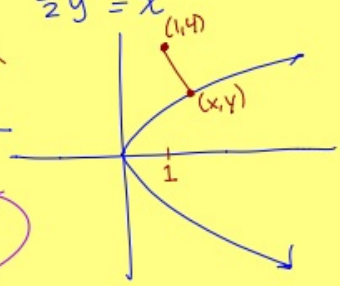
$L = 2400 - 2W$

$L = 2400 - 1200 = 1200$

the dimensions are 600ft × 1200ft (with the 1200ft side parallel to the river)

Example 3: Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.

$\frac{1}{2}y^2 = x$



Minimize the distance between $(1,4)$ & (x,y)

Primary EQ $d = \sqrt{(x-1)^2 + (y-4)^2}$

Secondary EQ: $y^2 = 2x$

$\frac{1}{2}y^2 = x$

$d = \sqrt{\left(\frac{1}{2}y^2 - 1\right)^2 + (y-4)^2}$

$D = d^2 = \left(\frac{1}{2}y^2 - 1\right)^2 + (y-4)^2$

$2y \cdot \frac{1}{2}y^2$

$D' = 2\left(\frac{1}{2}y^2 - 1\right)y + 2(y-4)$

$= y^3 - 2y + 2y - 8$

$= y^3 - 8$

D' $\begin{array}{ccc} - & & + \\ 0 & 2 & \end{array}$

$y^3 - 8 = 0$

$y^3 = 8$

$y = 2$

$x = \frac{1}{2}y^2$

$= \frac{1}{2}(2)^2 = 2$

$(2,2)$ is the point on the curve closest to $(1,4)$